Computer Organization and Assembly Language is a foundational subject in computer science that explores how computers operate at the hardware and low-level software level. It focuses on the internal structure of a computer system, including the CPU, memory hierarchy, input/output mechanisms, and data paths. This subject helps students understand how high-level code is translated into machine-level instructions through assembly language, which provides direct control over hardware. By learning assembly language, students gain insight into instruction sets, registers, addressing modes, and how operations like arithmetic, branching, and looping are handled by the processor. Overall, it bridges the gap between hardware and software, enabling a deeper understanding of system performance, efficiency, and architecture design.